

Element B: Documentation and Analysis of Prior Design Attempts

Study: https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.youtube.com%2Fwatch%3Fv%3De1FVSpkw6q4&psig=AOvVaw0S_EsMxPeTPBzzDhE6ulOJ&ust=1746714044150000&source=images&cd=vfe&opi=89978449&ved=0CBgQ3YkBahcKEwjQueSCx5GNAXUAAA_AAHQAAAAAQBA



Teachers have started using Arduino boards in class to help us learn about electronics and programming in a fun way. For example, this teacher worked on an "Arduino Tutorial: LED Sequential Control - Beginner Project," where we program multiple LEDs to light up in a sequence. While Arduino is great, Makey Makey circuit boards can be even better for kids and teachers. They let us turn everyday objects into touchpads, making it easy to create interactive projects without needing extensive coding knowledge. Makey Makey is more intuitive and encourages creativity, which is especially helpful for younger students. However, with both tools, we sometimes rely too much on online tutorials, which can limit our problem-solving skills, and getting started can still be a bit challenging for beginners.

Strengths:

- Arduino LED projects provide a practical way for students to learn about electronics, circuits, and programming, enhancing their understanding through hands-on experience.
- Arduino boards can be used for a wide range of projects, from simple LED control to complex robotics, allowing for creativity and exploration.
- Working with Arduino helps students develop valuable coding skills, as they learn to write and upload code to control the LEDs and other components.

Weaknesses:

- For beginners, especially younger students, the initial learning curve can be challenging, as it requires understanding both electronics and programming concepts.
- Some students may find Arduino's capabilities overwhelming for simple projects, which could lead to frustration rather than enjoyment.
- Students might become overly reliant on online tutorials, which can hinder their ability to think critically and solve problems independently.
- While Arduino boards themselves are affordable, additional components (like resistors, LEDs, and sensors) can add up, potentially making it less accessible for some classrooms.

As kids use the Makey Makey kits, they often encounter problems that challenge their creativity and problem-solving skills. For example, sometimes the materials we choose don't conduct electricity well, which can be frustrating when our circuits don't work. Figuring out why something isn't registering as a touch input can be tricky, but these challenges push us to think critically and collaborate to find solutions. It's all part of the learning process, and overcoming these obstacles makes our projects even more rewarding.

Study: <https://makeymakey.com/piano>

The Makey Makey Cardboard Piano is a fun and creative project that allows users to create a playable piano using simple materials like cardboard, aluminum foil, and the Makey Makey kit. This project transforms everyday objects into musical instruments, making it an engaging way to learn about circuits and sound production. This is one of the pianos made using Makey Makey



Strengths

- It requires minimal materials, making it easy for students to gather what they need and start building. The project encourages creativity, as students can design their pianos uniquely, choosing different shapes and colors for the keys.
- It provides a hands-on learning experience that reinforces concepts of conductivity and circuit design while allowing students to explore music and sound.

Weaknesses

- The piano's functionality can be limited by the materials used; for instance, if the conductive surfaces are not properly connected or if the materials do not conduct electricity well, it can lead to frustrating experiences.
- The sound quality may not be as high as that of a traditional piano, which could affect the overall musical experience.
- While the project is great for individual creativity, it may require guidance and support from teachers to help students troubleshoot issues and fully understand the underlying concepts.